

# Ehab Ashraf Mohamed

AI/ML Engineer

Military Service: Completed

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## Professional Summary

AI and Machine Learning Engineer with hands-on experience across the full ML lifecycle, from data engineering to model training and evaluation. Recently expanded into Generative AI, engineering an end-to-end Retrieval-Augmented Generation (RAG) Knowledge Assistant using LangChain, Hugging Face Transformers, and the ChromaDB vector database to deliver grounded, source-cited answers from a custom knowledge base. Skilled in prompt engineering, LLM integration, NLP, and applying generative AI to solve practical business problems, combined with a strong foundation in classical ML and a fast learning curve on new AI technologies. Currently expanding my skill set toward agentic AI — applications that can reason, plan, and take multi-step actions using tools — with LangGraph as my next focus area within the LangChain ecosystem.

## Technical Skills

- **Generative AI & LLMs:** LLMs, LangChain, Hugging Face Transformers, Prompt Engineering, Retrieval-Augmented Generation (RAG), Conversational AI Design, Open AI APIS
- **Deep Learning & NLP:** TensorFlow, Keras, Transformers, Neural Networks, NLP, Computer Vision, NLTK
- **Machine Learning:** scikit-learn, NumPy, Pandas, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Time Series Analysis, Reinforcement Learning
- **Vector Databases & Retrieval:** ChromaDB, FAISS, Sentence-Transformer Embeddings, Semantic Search, Similarity Search
- **MLOps & Tools:** MLflow, Git
- **Data Engineering & Visualization:** Web Scraping, Data Preprocessing, Data Mining, Advanced Analytics, Matplotlib, Seaborn
- **Cloud & Platforms:** AWS (foundational — AWS Academy certified), Hugging Face Hub
- **Tools & Platforms:** Jupyter Notebook, VS Code, GitHub, Kaggle

## Work Experience

### Machine Learning Engineer & Data Analyst | Quantum Bits Company | Mar 2024 – Mar 2025

- Engineered, deployed, and monitored stock price prediction models using financial, economic, and alternative data sources, improving prediction accuracy by 15% over baseline models.
- Trained and maintained 50+ individual stock-level models, generating a 10-year performance track record to identify and surface top-performing strategies.
- Analyzed quarterly financial peaks and global economic indicators, reducing forecasting error by 12% in quarterly revenue predictions.
- Built and optimized dynamic investment portfolios that outperformed the S&P 500 by 12% annually across 10-year backtests.
- Implemented MLOps practices (continuous training, deployment, and monitoring), cutting model update cycle time by 30%.
- Benchmarked model-based portfolios against market indices, achieving consistent alpha generation over multiple quarters.

### Machine Learning Engineer | Freelance (Remote) | 2022 – 2024

- Delivered end-to-end ML solutions spanning NLP, computer vision, and predictive analytics for clients, improving model performance by up to 18%.
- Applied regularization and hyperparameter tuning techniques, boosting model generalizability and reducing overfitting by 25%.
- Designed automated visualization dashboards that reduced client reporting time by 40%.

- Preprocessed and engineered features from datasets exceeding 100,000 records, improving downstream model efficiency.
- Deployed end-to-end ML models with monitoring pipelines, ensuring 99% uptime for client-facing applications.
- Produced detailed model performance reports (accuracy, precision, recall, F1-score) to support data-driven client decision-making.

## Projects

### AI Knowledge Assistant — Retrieval-Augmented Generation (RAG) Chatbot

- Architected an end-to-end conversational AI assistant using LangChain, Hugging Face Transformers, and ChromaDB to deliver grounded, source-attributed answers from a custom multi-domain knowledge base.
- Built a document ingestion and retrieval pipeline combining PDF and Wikipedia API sources across 15 AI/ML domains, using NLTK-based semantic chunking with a Hugging Face tokenizer to optimize retrieval quality.
- Generated dense vector embeddings (sentence-transformers/all-MiniLM-L6-v2) and implemented top-k similarity search over a persisted Chroma vector store to ground LLM responses in verified source content.
- Integrated a Qwen2.5-1.5B-Instruct LLM via a Hugging Face transformers pipeline with custom prompt engineering and conversation buffer memory, reducing hallucinations through controlled, context-grounded generation and explicit fallback handling for unanswerable questions.

### Portfolio Strategy Optimization (Quantum Bits)

- Designed and evaluated multiple quantitative investment strategies on a quarterly basis, dynamically adjusting portfolio allocation in response to changing market conditions to improve returns and reduce risk.

### Stock Price Prediction & Portfolio Analysis (Quantum Bits)

- Developed individual predictive models for 50+ stocks using 10 years of financial and economic data; benchmarked portfolio performance against the S&P 500 and analyzed quarterly market peaks and bottoms.

### English-to-Arabic Neural Machine Translation

- Implemented a Transformer-based neural machine translation model to translate text from English to Arabic, applying advanced deep learning techniques to achieve high translation accuracy.

### Customer Sentiment Classification (NLP)

- Applied NLP and machine learning techniques to classify customer sentiment in product reviews, improving businesses' ability to extract actionable insights from customer feedback.

## Education

### Bachelor of Computers and Artificial Intelligence | Benha University (Benha, Egypt) | Aug 2020 – Aug 2024

- Graduated with a cumulative GPA of 3.77/4.0 (A). Ranked 9th in the Artificial Intelligence Department and 10th in college.
- Core coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Robotics, Transformers, Advanced Data Analytics, and Data Mining.

## Certifications

- **Data Engineering** — AWS Academy (May 2024)
- **4th Place, Accessibility & Entertainment Track** — Emerging Technologies Hackathon, empowering people of determination (Feb 2024)

## Achievements

- Ranked 9th in the Artificial Intelligence Department, Benha University.
- Ranked 10th in college, Faculty of Computers and Artificial Intelligence.
- Won 4th place in a national hackathon competition (Accessibility & Entertainment track).

## Languages

Arabic (Native) | English (Professional Working Proficiency)